LECTURE 07 · REVIEW NOTES

What Is a Share Worth?

Book equity, market equity, discounted cash flows and dividend discount models.

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1. Begin with Citi and NVIDIA

The market price of a share is not the same as its book value. Book equity is the accounting residual after recorded liabilities are subtracted from recorded assets. Market equity is the price investors currently pay for all common shares outstanding. Compare common equity with common equity, using a consistent date and share basis.

Book equity = Assets − Liabilities

Market equity = Share price × Common shares outstanding

Historical snapshot	Book measure	Market measure	Market / book
Citi · Dec. 31, 2024	$101.62 per share	$70.39 per share	0.69×
NVIDIA · Jan. 26, 2025	$79.327bn common equity	$3,490.91bn market equity	44.0×

NVIDIA: 24.477 billion shares × $142.62 = $3,490.91 billion. The price is the Jan. 24, 2025 close, the last trading day before its fiscal close. Its assets were $111.601bn and liabilities $32.274bn. Citi and NVIDIA have different fiscal closing dates; these are labeled historical examples, not current quotations.

Below book does not automatically mean cheap. A discount can reflect low expected profitability relative to the required return, anticipated losses or other risks. **Above book does not automatically mean expensive.** Expected growth, competitive advantages and internally developed capabilities can support value that is not fully reflected in recorded net assets.

The central question is: **what future cash flows and risk assumptions would justify the market price?** Neither ratio proves that the market is right or wrong. These notes use fictional mature companies to practice DDM; NVIDIA's current dividend alone is not a sufficient one-stage valuation model.

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2. Three questions behind every DCF

A discounted cash flow model connects expected future payments to value today. Discounting reflects both the time value of money and the return required for the cash flows' risk. The model is an estimate conditional on its assumptions, not a promise of the future trading price.

$$\text{Value today} = CF_1/(1 + r) + CF_2/(1 + r)^2 + \ldots + CF_n/(1 + r)^n$$

Question	Equity valuation in this lecture
Which cash flows?	Dividends available to common shareholders.
Which discount rate?	Cost of equity k: the return shareholders require.
Which horizon?	One stable-growth stage, or explicit early dividends followed by stable growth.

Connecting Lecture 06 to valuation

$$k = r_f + \beta \times [E(r_M) - r_f]$$

If the risk-free rate is 4%, beta is 1.20 and the market risk premium is 5%, then k = 4% + 1.20 × 5% = **10%**. The premium is already the excess return; do not subtract the risk-free rate a second time.

Worked example: a finite stream

Year	Cash flow	Discounting at 10%	Present value
1	$2.00	2.00 / 1.10^1	$1.8182
2	$2.20	2.20 / 1.10^2	$1.8182
3	$2.40	2.40 / 1.10^3	$1.8032
Total			$5.4395

This sum values only the three displayed payments. If the business continues afterward, omitting later distributions understates its value. A terminal value represents those later payments.

Match the cash flow and the discount rate

Dividends go to shareholders, so use the cost of equity. Free cash flow to the firm goes to debt and equity providers, so an enterprise-value model uses WACC. Enterprise value and market equity are different measures. Do not discount dividends using WACC merely because it is called a cost of capital.

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3. Gordon growth and the sources of growth

For a mature dividend-paying firm, the Gordon model assumes dividends grow at a constant rate forever and the required return remains constant. D_0 is the dividend just paid; D_1 is next year's expected dividend.

$$D_1 = D_0(1 + g)$$
$$P_0 = D_1 / (k - g), \text{ with } k > g$$

Use decimals in the denominator: 10% − 4% is 0.10 − 0.04 = 0.06. The formula requires a positive spread k − g. If g approaches k, the valuation becomes extremely sensitive; g ≥ k does not produce a valid finite Gordon value.

Growth from reinvestment

$$\text{Retention } b = 1 - \text{payout ratio } \delta$$
$$g = b \times ROE$$
$$D_0 = EPS_0 \times \delta$$

This growth relation assumes stable profitability and reinvestment policies. ROE measures earnings relative to book equity. Reinvestment supports growth, but growth alone does not guarantee value creation: the return on the additional investment matters.

Worked example: growth, value and a trading signal

Input	Value
EPS_0; payout ratio	\$5.00; 60%
ROE; required return k	10%; 10%
Market quote	\$48.00

Step	Calculation
Retention	b = 1 − 0.60 = 0.40
Growth	g = 0.40 × 10% = 4%
Dividend just paid	D_0 = \$5.00 × 0.60 = \$3.00
Next dividend	D_1 = \$3.00 × 1.04 = \$3.12
Model value	P_0 = \$3.12 / (0.10 − 0.04) = \$52.00
Model-relative signal	\$52.00 > \$48.00: undervalued under these assumptions.

The implied upside is 52/48 − 1 = **8.33%**. This is a comparison between model and quote, not a guaranteed investment return. Challenge the growth and discount-rate assumptions before treating the signal as compelling.

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4. Same cash flows, different required return

A valuation can change even when the dividend forecast stays fixed. Higher required returns lower present value. Changes in risk-free rates, systematic risk or the market risk premium can all affect k. In the classroom experiment, only k changes; the market quote is deliberately held fixed to isolate the mechanism.

Classroom experiment: D_1 = \$3.00; g = 3%; market quote = \$55.00

Required return	Gordon value	Compared with \$55
8%	\$3 / (0.08 − 0.03) = \$60.00	Model above quote
9%	\$3 / (0.09 − 0.03) = \$50.00	Model below quote
10%	\$3 / (0.10 − 0.03) = \$42.86	Model below quote

A **one-percentage-point** increase in k, from 8% to 9%, lowers value by \$10 or **16.67%**. The dividend and growth forecast did not change. A model-relative buy signal became a sell signal at the unchanged \$55 quote.

Worked example from the lecture

The guided example uses D_0 = \$2.86, g = 3.07%, k = 6.46%, and a market price of \$75.77. Keep intermediate precision:

Step	Result
$D_1 = 2.86 \times 1.0307$	\$2.947802
$P_0 = D_1 / (0.0646 - 0.0307)$	\$86.9558
Implied $k = D_1 / 75.77 + 0.0307$	6.9605%

The market price can be reconciled with the same dividend-growth forecast by requiring about 6.96%, rather than 6.46%. That small difference is enough to explain the valuation gap. This inversion is useful: ask what the price implies before declaring it wrong.

Discussion habits

Predict direction first, calculate second. State what is held fixed. Distinguish percentage points from percentage changes. Finally, ask whether the new assumption is economically reasonable rather than choosing an assumption merely to obtain a preferred price.

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5. Two stages: value the transition and maturity

A young or rapidly changing business may not be well represented by a single perpetual growth rate. A two-stage DDM forecasts early dividends explicitly and applies stable growth only after that transition. In Activity 2, the first three dividends are given and stable growth begins with year 4.

$$TV_3 = D_4 / (k - g_S) = D_3(1 + g_S) / (k - g_S)$$
$$P_0 = D_1/(1+k) + D_2/(1+k)^2 + D_3/(1+k)^3 + TV_3/(1+k)^3$$

Timing matters: TV_3 is a value at year 3, immediately after the third dividend. It summarizes dividends from year 4 onward. It is not a cash flow at time 0 and must be discounted three years. D_3 belongs in the explicit stage and is not the numerator of TV_3 without first growing it to D_4.

Worked example: independent practice case

Assume D_1 = \$2.00, D_2 = \$2.40, D_3 = \$2.88, k = 10% and stable growth g_S = 3%. These are practice inputs, not an individual graded case.

Step	Calculation	Result
First stable dividend	$D_4 = 2.88 \times 1.03$	\$2.9664
Value at year 3	$TV_3 = D_4 / (0.10 - 0.03)$	\$42.3771
Year 1 dividend	2.00 / 1.10	\$1.8182
Year 2 dividend	$2.40 / 1.10^2$	\$1.9835
Year 3 dividend	$2.88 / 1.10^3$	\$2.1638
PV of terminal value	$TV_3 / 1.10^3$	\$31.8386
Value today	Sum of the four present values	\$37.8040

The share is worth **\$37.80** under these assumptions. Notice that \$42.38 is the future terminal value, not today's total share value. Rounding each intermediate value too early can distort the final answer.

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6. Why terminal value deserves scrutiny

In the practice case, PV(TV) is \$31.8386 of a total value of \$37.8040: **84.22%**. This is not automatically a modeling mistake. A continuing firm may pay dividends for decades after a short explicit forecast. But it means long-run assumptions deserve close attention.

Terminal-value share = $PV(TV_n) / P_0$
P_0 = PV(explicit dividends) + PV(terminal value)

Scenario	k	Stable g	P_0	$PV(TV) / P_0$
Base case	10%	3%	\$37.80	84.2%
Higher required return	11%	3%	\$32.97	82.2%
Higher stable growth	10%	4%	\$43.47	86.3%

Only the stated rate changes; the first three dividends remain fixed. Raising k lowers the present value of both the explicit dividends and the terminal value. Raising stable growth changes D_4 and narrows $k - g_S$, increasing terminal value through two channels.

Reading the classroom result charts

Activity 1: each bar compares the calculated value of a student's own case with its own market quote. A positive percentage gap is a model-relative buy signal; a negative gap is a sell signal. Comparing raw dollar values across different companies is less informative.

Activity 2: the green component is PV of the explicit dividends; the gold component is PV of terminal value. Their sum is P_0. Compare terminal-value shares and discuss which cases depend most heavily on the long run.

Avoid these common errors

Error	Correction
Using g = ROE	Multiply ROE by retention, not payout.
Putting percentages directly in a denominator	Convert 10% to 0.10 and 3% to 0.03.
Using D_0 in Gordon's numerator	Grow it once to obtain D_1.
Using TV_3 as today's value	Discount TV_3 and add early-dividend PVs.
Treating $k \le g$ as a meaningful negative valuation	Reassess the perpetual-growth assumptions.
Interpreting model upside as a promised return	The signal depends on forecasts and required returns.

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7. Activities and a final self-check

The two graded activities total **5 points**. Each student has a distinct, repeatable set of fictional-company inputs. The method is shared; copying another case's numerical answers will not solve your case.

Activity	Required outputs	Points
1 · Gordon + trading signal	g, k, P_0, and undervalued/buy or overvalued/sell	3
2 · Two-stage valuation	TV_3 and P_0	2

For Activity 1, the stated tolerance is ±0.05 percentage points for g and k and ±1% for P_0. Activity 2 permits ±1% for TV_3 and P_0. Keep at least four decimals in intermediate calculations. Complete Activity 1 before Activity 2.

Only correct, saved answers enter the instructor's discussion charts. Attempts, participation and scores are separate records. Instructor previews and rate-shock demonstrations do not submit on a student's behalf or change a score. In Review mode, graded submissions are closed.

Self-check: answer before reading the solutions

1. Book value is $100 per share and market price is $70. Is the stock necessarily undervalued?
2. D_0 = $2, g = 4%, k = 9%. What is P_0?
3. The same firm's k rises to 10%, with dividends and growth unchanged. What happens?
4. A terminal value is calculated at year 3. How many years should it be discounted?

Solutions. (1) No; the book-to-market gap alone does not identify mispricing. (2) D_1 = $2.08; P_0 = 2.08/(0.09 − 0.04) = **$41.60**. (3) P_0 = 2.08/(0.10 − 0.04) = **$34.67**, a 16.67% decrease. (4) Three years, adding separately the PV of dividends in years 1–3.

Sources and further review

Lecture readings: Bodie, Kane and Marcus, Chapters 12.7, 13.2 and 13.3. Discover, the cash-flow-model comparison, the life-cycle illustration and an extra worked example are optional lecture appendices.

Citigroup: Fourth Quarter and Full Year 2024 Results - book value per common share.
NVIDIA: FY2025 Form 10-K - balance sheet and common shares.
Citi historical closing prices and NVIDIA historical closing prices.
The local lecture provides a CSV with each of NVIDIA's 20 quarterly observations and filing sources. Prices and share counts use a consistent split-adjusted basis; dividends are not included in the price series.